- *Suppose $F(x)$ is the distribution function of a continuous random variable X. Then the probability density function $f(x)$ is given by*

$$f(x) = \frac{dF(x)}{dx} = F'(x)$$ *, whenever derivative exists.*

- *Suppose X is a random variable with probability mass or density function $f(x)$* ***The expected value or mean or mathematical expectation of X,*** denoted by $E(x)$ or μ *is*

$$E(X) = \begin{cases} \sum_x xf(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)\,dx & \text{if } X \text{ is continuous} \end{cases}$$

- *The* ***variance*** *of the random variable X* denoted by $V(X)$ or σ^2 (or σ_x^2) is

$$V(X) = E(X - E(X))^2 = E(X - \mu)^2$$

Properties of Mathematical expectation and variance

(i) $E(aX + b) = aE(X) + b$, where a and b are constants

Corollary 1: $E(aX) = aE(X)$	(when $b = 0$)
Corollary 2: $E(b) = b$	(when $a = 0$)

(ii) $Var(X) = E(X^2) - (E(X))^2$

(iii) $Var(aX + b) = a^2 Var(X)$ where a and b are constants

Corollary 3: $V(aX) = a^2V(X)$	(when $b = 0$)
Corollary 4: $V(b) = 0$	(when $a = 0$)

- Let X be a random variable associated with a Bernoulli trial by defining it as X (success) = 1 and X (failure) = 0, such that

$$f(x) = \begin{cases} p & x = 1 \\ q = 1 - p & x = 0 \end{cases} \quad \text{where } 0 < p < 1$$

- X is called a Bernoulli random variable and $f(x)$ is called the Bernoulli distribution.

- *If X* is a Bernoulli's random variable which follows Bernoulli distribution with parameter p, the mean μ and variance σ^2 are

$$\mu = p \quad \text{and} \quad \sigma^2 = pq$$

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